

● MEDIUM

● ADVANCED MATH

● ANSWER KEY

Nonlinear functions

07

10 answers · Rationale-rich · Teach from doc

Q1**C****C.** $\frac{4}{21}$

Choice C is correct. It's given that function f is defined by the equation $fx = 4x^{-1}$. The value of $f21$ is the value of fx when $x = 21$. Substituting 21 for x in the given equation yields $f21 = 421^{-1}$, which is equivalent to $f21 = 4\frac{1}{21}$, or $f21 = \frac{4}{21}$.

Choice A is incorrect. This is the value of $f21$ when $fx = -4x$, rather than $fx = 4x^{-1}$.

Choice B is incorrect. This is the value of $f21$ when $fx = 4x^{-1}$, rather than $fx = 4x^{-1}$.

Choice D is incorrect. This is the value of $f21$ when $fx = 4^{-1}x$, rather than $fx = 4x^{-1}$.

Q2**A****A.** $f(0)$

Choice A is correct. A y-intercept is the point in the xy-plane where the graph of the function crosses the y-axis, which is where $x = 0$. It's given that the y-coordinate of the y-intercept of the graph of function f is c . It follows that the coordinate pair representing the y-intercept must be $(0, c)$. Therefore, c must equal $f(0)$.

Choices B, C, and D are incorrect because $f(1)$, $f(2)$, and $f(3)$ would represent the y-value of the coordinate where $x = 1$, $x = 2$, and $x = 3$, respectively.

Q3**A****A. -12,0**

Choice A is correct. It's given that $y = fx$. The x -intercepts of a graph in the xy -plane are the points where $y = 0$. Thus, for an x -intercept of the graph of function f , $0 = fx$. Substituting 0 for fx in the equation $fx = x^2 - 18x - 360$ yields $0 = x^2 - 18x - 360$. Factoring the right-hand side of this equation yields $0 = x + 12x - 30$. By the zero product property, $x + 12 = 0$ and $x - 30 = 0$. Subtracting 12 from both sides of the equation $x + 12 = 0$ yields $x = -12$. Adding 30 to both sides of the equation $x - 30 = 0$ yields $x = 30$. Therefore, the x -intercepts of the graph of $y = fx$ are $-12, 0$ and $30, 0$. Of these two x -intercepts, only $-12, 0$ is given as a choice.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4**B**

- B. This toy rocket reaches an estimated maximum height of 502 feet 5.6 seconds after it is launched into the air.

Choice B is correct. The vertex of the graph of a quadratic equation is where it reaches its minimum or maximum value. When a quadratic equation is written in the form $y = a(x - h)^2 + k$, the vertex of the parabola represented by the equation is $(x, y) = (h, k)$. In the given equation $y = -16(x - 5.6)^2 + 502$, the value of h is 5.6 and the value of k is 502. It follows that the vertex of the graph of this equation in the xy -plane is $(x, y) = (5.6, 502)$. Additionally, since $a = -16$ in the given equation, the graph of the quadratic equation opens down, and the vertex represents a maximum. It's given that the value of y represents the estimated height, in feet, of the toy rocket x seconds after it is launched into the air. Therefore, this toy rocket reaches an estimated maximum height of 502 feet 5.6 seconds after it is launched into the air.

Choice A is incorrect. The 16 in the equation is an indicator of how narrow the graph of the equation is rather than where it reaches its maximum.

Choice C is incorrect. The 16 in the equation is an indicator of how narrow the graph of the equation is rather than where it reaches its maximum.

Choice D is incorrect. This is an interpretation of the vertex of the graph of the equation $y = -16(x - 502)^2 + 5.6$, not of the equation $y = -16(x - 5.6)^2 + 502$.

Q5**11/4**

Accept also: 2.75

The correct answer is $\frac{11}{4}$. It's given that the function f is defined by $f(x) = 5\frac{1}{4} - x^2 + \frac{11}{4}$.

Substituting $\frac{1}{4}$ for x in this equation yields $f\left(\frac{1}{4}\right) = 5\frac{1}{4} - \left(\frac{1}{4}\right)^2 + \frac{11}{4}$, which is equivalent $f\left(\frac{1}{4}\right) = 50^2 + \frac{11}{4}$, or $f\left(\frac{1}{4}\right) = \frac{11}{4}$. Therefore, the value of $f\left(\frac{1}{4}\right)$ is $\frac{11}{4}$. Note that 11/4 or 2.75 are examples of ways to enter a correct answer.

Q6**C****C. Three**

Choice C is correct. If a value of x satisfies $f(x) = 0$, the graph of $y = f(x)$ will contain a point $(x, 0)$ and thus touch the x -axis. Since there are 3 points at which this graph touches the x -axis, there are 3 values of x for which $f(x) = 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

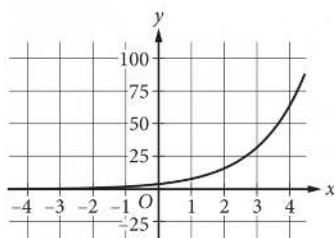
Q7**B****B. 19**

Choice B is correct. The area of a rectangle is equal to its length multiplied by its width. Multiplying the given length, x units, by the given width, $x - 15$ units, yields $x(x - 15)$ square units. If the rectangle has an area of 76 square units, it follows that $x(x - 15) = 76$, or $x^2 - 15x = 76$. Subtracting 76 from both sides of this equation yields $x^2 - 15x - 76 = 0$. Factoring the left-hand side of this equation yields $(x - 19)(x + 4) = 0$. Applying the zero product property to this equation yields two solutions: $x = 19$ and $x = -4$. Since x is the rectangle's length, in units, which must be positive, the value of x is 19.

Choice A is incorrect. This is the width, in units, of the rectangle, not the value of x .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the area, in square units, of the rectangle, not the value of x .

Q8**D****D.**

Choice D is correct. The y-intercept of the graph of an equation is the point $(0, b)$, where b is the value of y when $x = 0$. For the given equation, $y = 4$ when $x = 0$. It follows that the y-intercept of the graph of the given equation is $(0, 4)$. Additionally, for the given equation, the value of y doubles for each increase of 1 in the value of x . Therefore, the graph contains the points $(1, 8)$, $(2, 16)$, $(3, 32)$, and $(4, 64)$. Only the graph shown in choice D passes through these points.

Choices A and B are incorrect because these are graphs of decreasing, not increasing, exponential functions. Choice C is incorrect because the value of y increases by a growth factor greater than 2 for each increase of 1 in the value of x .

Q9**A**

A. The height of the object 1.4 seconds after being thrown straight up in the air is 1.64 feet.

Choice A is correct. The value 1.4 is the value of x , which represents the number of seconds after the object was thrown straight up in the air. When the function h is evaluated for $x = 1.4$, the function has a value of 1.64, which is the height, in feet, of the object.

Choices B and C are incorrect and may result from misidentifying seconds as feet and feet as seconds. Additionally, choice C may result from incorrectly including the initial height of the object as the input x . Choice D is incorrect and may result from misidentifying height as speed.

Q10

A

A. $A_t = 36,100.001.05^t$

Choice A is correct. Since it's given that the account balance, A_t , in dollars, after t years can be modeled by an exponential function, it follows that function A can be written in the form $A_t = Nr^t$, where N is the initial value of the function and r is a constant related to the growth of the function. It's given that the initial balance of the account is \$ 36,100.00, so it follows that the initial value of the function, or N , must be 36,100.00. Substituting 36,100.00 for N in the equation $A_t = Nr^t$ yields $A_t = 36,100.00r^t$. It's given that the account balance after 13 years, or when $t = 13$, is \$ 68,071.93. It follows that $A_{13} = 68,071.93$, or $36,100.00r^{13} = 68,071.93$. Dividing each side of the equation $36,100.00r^{13} = 68,071.93$ by 36,100.00 yields $r^{13} = \frac{68,071.93}{36,100.00}$. Taking the 13th root of both sides of this equation yields $r = \sqrt[13]{\frac{68,071.93}{36,100.00}}$, or r is approximately equal to 1.05. Substituting 1.05 for r in the equation $A_t = 36,100.00r^t$ yields $A_t = 36,100.001.05^t$, so the equation $A_t = 36,100.001.05^t$ could define A .

Choice B is incorrect. Substituting 0 for t in this function indicates an initial balance of \$ 31,971.93, rather than \$ 36,100.00.

Choice C is incorrect. Substituting 0 for t in this function indicates an initial balance of \$ 31,971.93, rather than \$ 36,100.00. Additionally, this function indicates the account balance is decreasing, rather than increasing, over time.

Choice D is incorrect. This function indicates the account balance is decreasing, rather than increasing, over time.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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